Revitalizing Filipino Heritage Through Augmented Reality



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Tinikling is a dance where people smile



Tinikling is derived from the Tikling bird as it hops, spins and dances through traps set by rice farmers



[https://www.kollectivehustle.com/blog/origin-of-the-tinikling-dance]



Carries Tikling bird’s
playful and
mischevious energy



Represents
agricultural lifestyle

Dance is the physical architecture of community and a living shield against cultural erasure



[https://www.islandtribe.co/blogs/news/the-tinikling-dance]

“Our folk dances are the living records of our race”

Published the Philippine National Dances (1946)

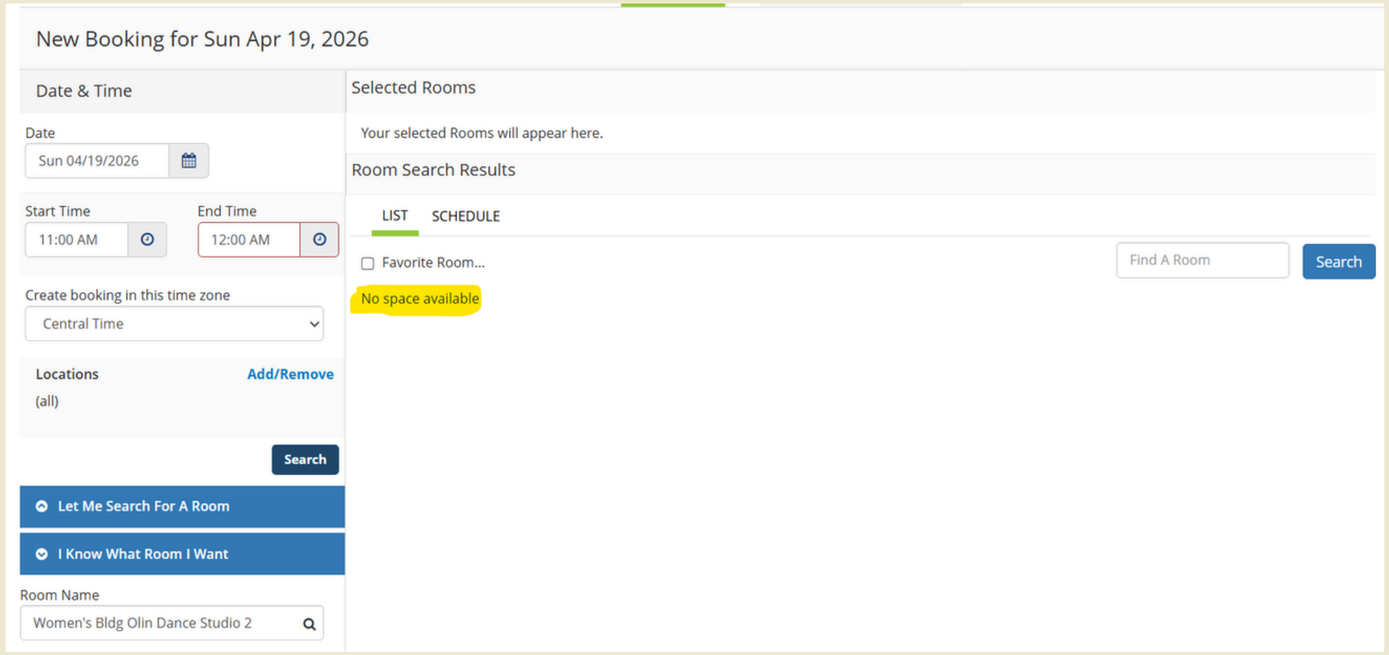
Wanted to Protect Filipino Folk Dances from Western Modernization

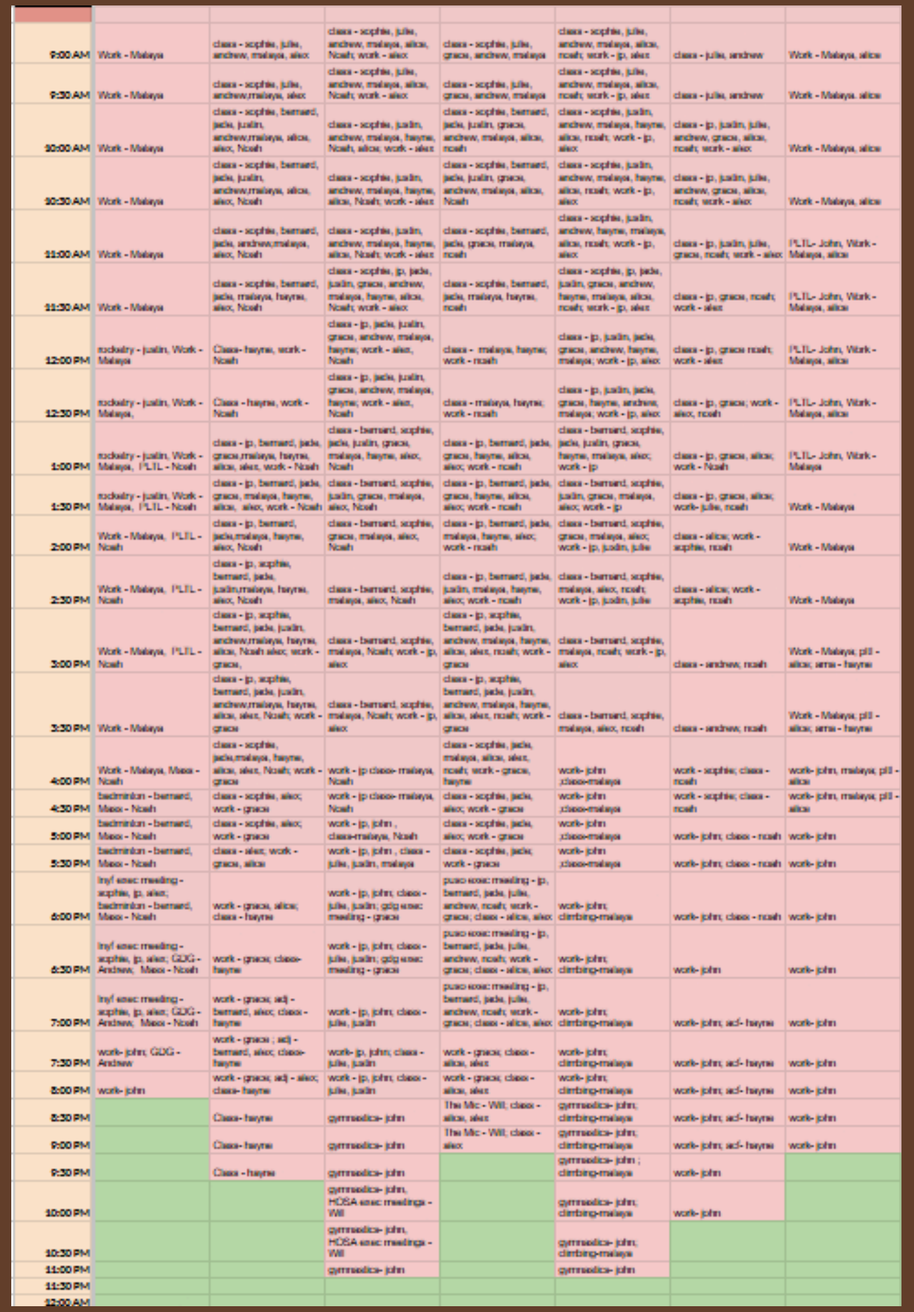


She Failed.



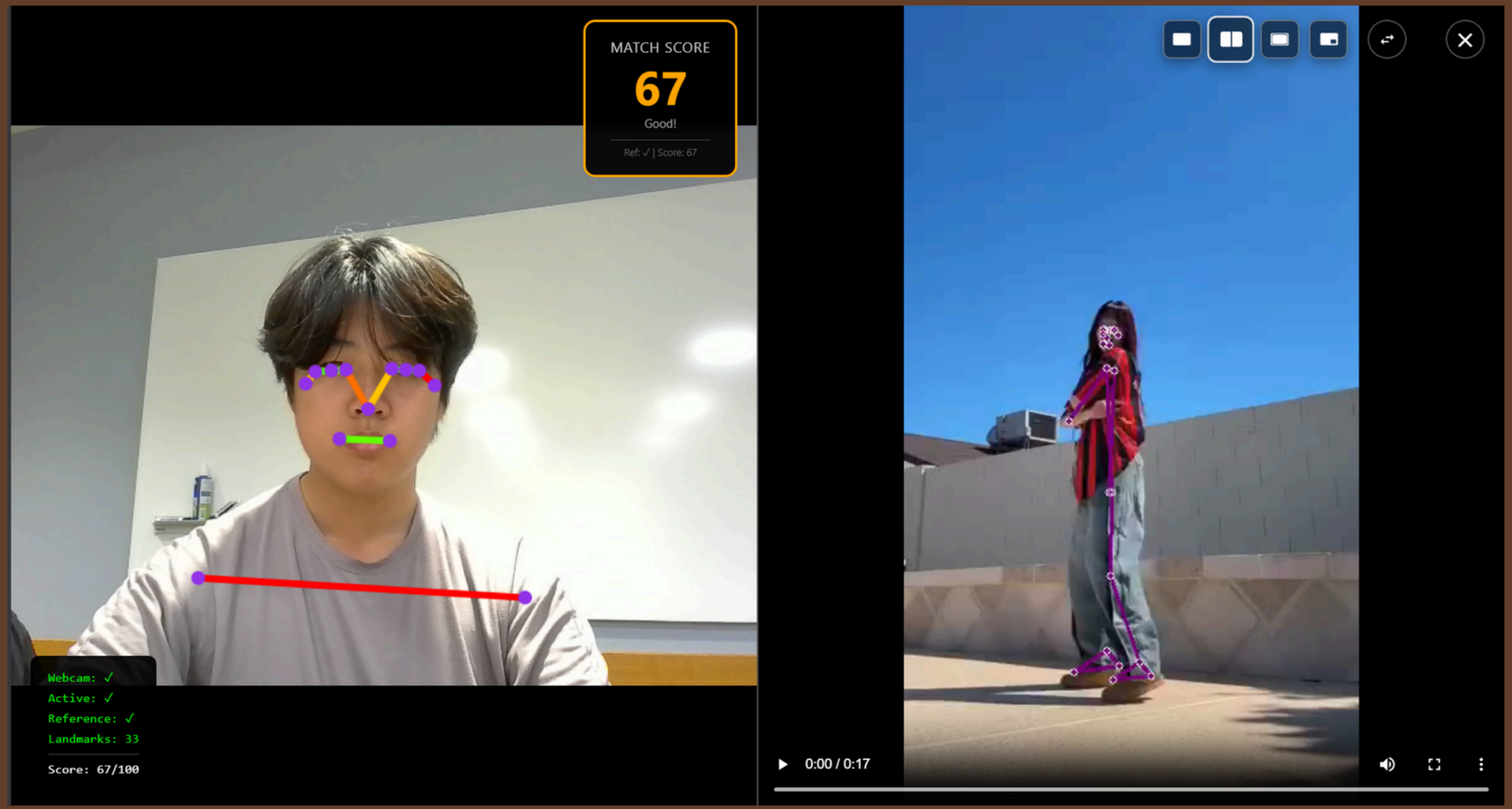
Physical barriers—like logistics, time, and space—are the primary drivers of cultural erosion





So... How do we fix it?

Augmented Reality (AR) and Pose Estimation can act as a digital sanctuary, allowing for Tinikling anywhere and anytime



The Architecture

Camera

60 frames per second (fps) video capture

GPU

Resize & convert to tensor

NPU

Pose estimation model

CPU

Collision detection feet vs bamboo poles

User Interface

Draw AR poles, vibrations on collision

Latency is the tightest bottleneck, but it still will not be an issue

We must do all tasks before next frame loads

$\Rightarrow$ 1000 ms / 60 frames $\cong$ 16.67 ms / frame

Conclusion

Presentation template designed by Canva Creative Studio



THANK YOU!

Questions?